

COMPUTER AIDED MACHINE DRAWING LAB

II B.TECH II Semester: MECHANICALENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME21	PCC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	40	60	100
COURSE OVERVIEW: The purpose of this course is to introduce conventional representation of materials and machine components, drawing of various machine elements, joints as well as assembly drawing of engine parts and machine parts through Computer Aided Drafting. Machine drawing can be used as a communicating medium in industries to furnish all details needed for the manufacturing and assembling of a machine. COURSE OUTCOMES: The course should enable the students to: 1. Understand and apply Indian and international standards while drawing machine components. 2. Illustrate various machine elements and simple parts in drawings. 3. Construct the drawing of riveted joints, couplings and bearings. 4. Discover the engine part drawings for apt assembled views. 5. Select the elements of machine parts for appropriate assembled views.								
UNIT-I	MACHINE DRAWING CONVENTIONS							
Need for drawing conventions – Introduction to standard conventions: 1. Types of Drawings – Introduction, Drawing area, Title block size, location and details. 2. Conventional representation of materials and machine elements. 3. Methods of dimensioning, representation of geometric dimensioning & tolerance on drawings. 4. Types of sections – drawing of full and half section.								
UNIT-II	DRAWING OF MACHINE ELEMENTS AND SIMPLE PARTS							
Selection of Views, additional views for the following machine elements and parts with every drawing proportions. Popular forms of screw threads, bolts, nuts, set screws, keys, cotter joint and knuckle joint.								
UNIT-III	RIVTED JOINTS, COUPLINGS AND BEARINGS							
Riveted joints for plates, flanged coupling, universal coupling. Journal, pivot, collar and foot step bearings.								
UNIT-IV	ASSEMBLED VIEWS OF ENGINE PARTS							
Drawing of assembled views for the part drawings of the following using conventions. Engine parts – Stuffing box, eccentric, connecting rod, piston.								
UNIT-V	ASSEMBLED VIEWS OF MACHINE PARTS							
Drawing of assembled views for the part drawings of the following using conventions. Machine parts - Screws jack, drill jig, Plummer block, clapper block.								

Text Books:

1. Machine Drawing – K.L. Narayana, P. Kannaiah & K. Venkata Reddy / New Age Publishers.
2. Machine Drawing with AutoCAD – Goutham Pohit, Goutham Ghosh / Pearson.
3. Machine Drawing – N. Sidheswar, P. Kanniah and V.V.S. Sastry / Tata McGraw Hill.

Reference Books:

1. Machine Drawing – N.D. Bhatt / Charotar Publishing House.
2. Machine Drawing – P.S. Gill / S.K. Kataria & Sons.
3. Machine Drawing – R.K. Dhawan / S.Chand Publications
4. IS code SP 46(2003): Engineering Drawing Practice for schools and colleges by Bureau of Indian Standards.
5. Bureau of Indian Standards